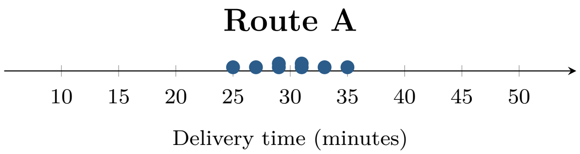
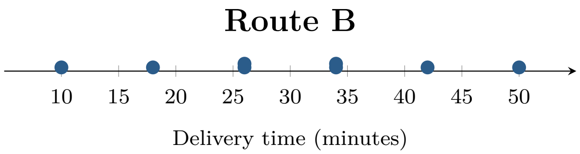
Same Mean, Different Deadline Risk

Topic 4.8 · TODAY'S PROBLEM

Suppose a dispatcher models delivery time using eight equally likely outcomes for each route, shown in the dotplots. An analyst reports that both routes have mean 30 minutes, with $\sigma_A = 3$ minutes and $\sigma_B = 12$ minutes.

The dispatcher says, *The routes have the same mean, so it does not matter which one a customer gets.* A customer has 40 minutes before an appointment and places a high cost on being late.





FIRST TAKE (5 min, on your sheet)

Which route would you assign to this customer? Cite one number or dotplot feature.

DOK 1 **(a)** For each route, state its minimum and maximum modeled time and count the outcomes longer than 40 minutes.

DOK 2 **(b)** Verify the reported mean and population standard deviation for each route from the eight equally likely outcomes. Show the sums of squared deviations.

DOK 3 **(c)** Decide which route better matches this customer's priority. Cite the parameter and dotplot feature that settle your choice, and explain what the dispatcher's mean-only argument would mislead a reader into believing. Then suppose every delivery gets 5 minutes of loading time: state both transformed means and standard deviations and decide whether your route choice changes.

Video + follow-along: `u4_lesson7-8_live.html` (scan the code)

Finish (a)–(c) after the video. Turn in your sheet.